

A high-angle, top-down photograph of a person wearing a bright yellow sweater, leaning over a black portable refrigerator. The refrigerator is open, revealing its interior which is packed with various items: several green cans (likely beer), a yellow bell pepper, a green bell pepper, a red bell pepper, some packaged snacks, and a bunch of green leafy vegetables. The refrigerator is sitting on a dark, textured surface, possibly the back of a vehicle. To the left of the refrigerator, there is a light blue portable fuel tank with two black caps. The background is a dry, brown, grassy field. The overall lighting is bright, suggesting it's daytime.

PORTABLE REFRIGERATOR TRAINING

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TOPICS

- Current Product Range
- Older Product Range
- Recon CF47DZW
- Fault Codes
- Compressor Start Failure
- Battery Protection Setting
- 619 Config
- Temperature Calibration
- Faulty Thermistor Symptoms
- Thermistor Resistances
- Emergency Override





Recon CF

Model	SKU
CF47DZW	97000133317



CFX5 Series

Model	SKU
CFX525	9620015957
CFX535	9620015958
CFX545	9620015959
CFX555	9620015960
CFX555IM	9620015961
CFX575DZ	9620015962
CFX595DZ	9620015963



CFX2 Series

Model	SKU OCEAN	SKU MOSS
CFX228	9620016988	97000050786
CFX237	9620016989	97000050787
CFX245	9620015892	97000050788
CFX257	9620015897	97000050789
CFX275DZ	97000150790	97000150839
CFX295DZ	97000150791	97000150840



CFF Series

Model	SKU
CFF12	CFF12
CFF20	CFF20
CFF35	CFF35
CFF45	CFF45
CFF50LPDZ	97000050792
CFF70DZ	CFF70DZ

DOMETIC CURRENT PRODUCT RANGE



CFX3 Series

Model
CFX325
CFX335
CFX345
CFX355
CFX355IM
CFX375DZ
CFX395DZ



CFX Series

Note: CFX series has many different versions from over the years such as: CFX, CFXW, CFX-, CFX2

Model
CFX18
CFX25
CFX35
CFX40
CFX50
CFX60
CFX80
CFX100



CF Series

Note: CF series has many different versions from over the years such as: CF, CF-, CF-ACVERA, CF-ACVERB, CF-AC, CF-OLD, CF-7LED, CF PACK, CF-DZ.

Model
CF18
CF25
CF35
CF40
CF50
CF60
CF80
CF110



CDF Series

Model
CDF11
CDF25
CDF35
CDF45

OLDER PRODUCT RANGE

Spare parts discontinued on majority of older models

Recon

CF47DZW

97000133317

Key Features

Dual Zone Cooling

Option to add Removable Divider and lid with either side of the divider able to run as a fridge or freezer

Removable Rechargeable Battery

Remove battery from the fridge for easy charging access

Multiple Charging Options

USB-C, USB-A and XT60 ports are built into the power setup,

Built In Solar Converter

MPPT solar converter for flexible charging

Glide Handle And Wheels

For easy moveability



Recon CF47DZW

SKU	97000133317
Connection voltage	12 – 24v DC 240v AC
Rated Current DC	12v = 10.5A 24V = 5.2A
Rated Current AC	240v = 0.58A
Cooling Capacity	-22°C to 20°C
Volume	47 L
Refrigerant	R600a (15g)
Battery Type	LithiumIronPhosphateLiFePO4
Rated Battery Voltage	12.8V DC
Battery End Of Discharge Voltage	11V DC
Battery Capacity	20 Ah

RECON CF47DZW FAULT CODES

Error Code	Fault	Possible Cause	Suggested Remedy
!ERROR 01	NTC open circuit SZ & DZ LARGE	Internal fault	Authorized repair centre inspection required
!ERROR 02	NTC short circuit SZ & DZ LARGE		
!ERROR 11	NTC open circuit DZ SMALL		
!ERROR 12	NTC short circuit DZ SMALL		
!ERROR 03	Communication error		Power off the device for at least 1 min, then restart. If the problem persists, authorized repair centre inspection is required.
!ERROR 09	Valve faulty		
!ERROR 32	Fan faulty		Power off the device for at least 5 min, then restart. If the problem persists, authorized repair centre inspection is required.
!ERROR 33	Compressor start fail		
!ERROR 34	Compressor speed low	• High ambient temperature • Air vent blocked or insufficient clearance • Internal fault	• Power off the device for at least 30 min or move to a cooler location, then restart. • Clear the vents. • Authorized repair center inspection required.
!ERROR 20	Battery fault	Internal fault	Read the Recon battery operating manual (Related Documents, page 3).
Alert Message	Description	Possible Cause	Suggested Remedy
Lid open	Lid open more than 3 min	Lid not fully latched	Ensure the lid is latched completely.
		Internal fault	Inspection by an authorized repair center required.
Low DC voltage	The device has stopped working on DC due to low voltage of external power supply	Insufficient supply voltage	Check voltage drop from battery to power outlet and increase wiring gauge if necessary.
		Battery voltage is too low	Test the battery and charge as needed.
		Battery protection is set too high	Select a lower battery protection setting.
High DC voltage	The device has stopped working on DC due to abnormally high voltage (> 31.5 V) of external power supply	Wrong DC supply voltage specification	Ensure DC supply voltage specification matches the product rating label.
		Faulty supply voltage	Check DC voltage supply and repair or replace if faulty.
		Internal fault	Inspection by an authorized repair center required.
Controller overheat	Controller over-temperature	High ambient temperature	- Power off the device for at least 30 min or move to a cooler location
		Air vent blocked or insufficient clearance	- Restart.
		Internal fault	- Clear the vents and restart. Authorized repair centre inspection required.
Internal battery low SOC	SOC of battery is low, but not reaching 0% yet	SOC of battery is low	Charge the battery.
Internal battery high temp	Recon battery over-temperature	Over-temperature protection of the battery triggered	Stop charging or discharging the device until it has cooled down.

CF & CFX FAULT CODES

If Digital Display Shows	Fault	
ER1	Thermistor circuit closed	1) Check NTC connection 2) If connection is ok, replace NTC
ER2	Thermistor circuit open	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module

Number Of Red Led Flashes	Fault	
1	Battery protection cut-out (The voltage is outside the cut-out setting).	1) Check supply Voltage under load, lead and wiring 2) If supply voltage is ok, check internal wiring and plugs 3) if wiring and plugs are ok, replace compressor control module
2	Fan over-current cut-out (The fan loads the electronic unit with more than 1Apeak).	1) Check fan wiring for short circuit, replace fan 2) If fan is ok, replace compressor control module
3	Motor start error (The rotor is blocked or the differential pressure in the refrigeration system is too high (>5 bar)).	1) Check power supply, lead and wiring 2) If power supply is ok, Check internal electrical connections 3) If electrical connections are ok, Check compressor windings 4) If compressor is ok, Change compressor control module
4	Minimum motor speed error (If the refrigeration system is too heavily loaded, the motor cannot maintain minimum speed 1,850 rpm).	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
5	Thermal cut-out of electronic unit (If the refrigeration system has been too heavily loaded or if the ambient temperature is high, the electronic unit will run too hot).	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart

CFF FAULT CODES

If Digital Display Shows	Fault	Process
Err 01 (SZ) / Err 01 (DZ)	NTC open circuit SZ & DZ LARGE	1) Check NTC connection 2) If connection is ok, replace NTC
Err 01 (SZ) / Err 11 (DZ)	NTC open circuit DZ SMALL	1) Check NTC connection 2) If connection is ok, replace NTC
Err 02 (SZ) / Err 02 (DZ)	NTC short circuit SZ & DZ LARGE	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module
Err 12 (SZ) / Err 12 (DZ)	NTC short circuit DZ SMALL	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module
Err 09 (DZ)	DZ valve faulty Icemaker valve faulty	1) Check if error disappears automatically within 30 seconds (can be false triggered by zone switching at the same time as error detection) 2) Check connections at valve and compressor control. 3) If connections are ok, check cooling performance of each zone, 4) If at least one zone is not cooling, activate valve and check if valve is receiving power. 5) If valve is not receiving power, replace module. 6) If valve is receiving power, visually check for blockages 7) If no sign of blockages, replace valve.
Err 03 (SZ) / Err 03 (DZ)	Communication error Between display and controller	1) Check connections at compressor control module and Display PCB 2) If connections are ok, Replace display PCB 3) If Replacing the display PCB hasn't fixed the issue, replace Compressor control module

Number Of Red Led Flashes	Fault	Process
2	Fan over-current	1) Check fan wiring for short circuit, replace fan 2) If fan is ok, replace compressor control module
3	Compressor start fail	1) Check power supply, lead and wiring 2) If power supply is ok, Check internal electrical connections 3) If electrical connections are ok, Check compressor windings 4) If compressor is ok, Change compressor control module
4	Compressor speed low	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
5	Controller over-temperature	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart

CFX3 & CFX5 FAULT CODES

Display	Description	Process
! Warning 32	Fan over current	1) Check fan wiring for short circuit, replace fan 2) If fan is ok, replace compressor control module
! Warning 33	Compressor start failure	1) Check power supply, lead and wiring 2) If power supply is ok, Check internal electrical connections 3) If electrical connections are ok, Check compressor windings 4) If compressor is ok, Change compressor control module
! Warning 34	Compressor speed low	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
! Warning 35	Controller over temperature	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
! Warning 01	NTC open circuit SZ & DZ Large	1) Check NTC connection 2) If connection is ok, replace NTC
! Warning 11	NTC open circuit DZ small	1) Check NTC connection 2) If connection is ok, replace NTC
! Warning 02	NTC Short circuit SZ & DZ Large	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module
! Warning 12	NTC short circuit DZ small	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module
! Warning 09	DZ valve faulty or Icemaker valve faulty	1) Check if error disappears automatically within 30 seconds (can be false triggered by zone switching at the same time as error detection) 2) Check connections at valve and compressor control. 3) If connections are ok, check cooling performance of each zone, 4) If at least one zone is not cooling, activate valve and check if valve is receiving power. 5) If valve is not receiving power, replace module. 6) If valve is receiving power, visually check for blockages 7) If no sign of blockages, replace valve.
! Warning 03	Communication error- Between display and controller	1) Check connections at compressor control module and Display PCB 2) If connections are ok, Replace display PCB 3) If Replacing the display PCB hasn't fixed the issue, replace Compressor control module
! Warning 43	Communication - Faulty display components	1) Replace display PCB

CFX2 FAULT CODES

Display	Description	Process
E 01	Low voltage / high voltage	1) Check supply Voltage under load, lead and wiring 2) If supply voltage is ok, check internal wiring and plugs 3) if wiring and plugs are ok, replace compressor control module
E 02	Fan over-current	1) Check fan wiring for short circuit, replace fan 2) If fan is ok, replace compressor control module
E 03	Compressor start failure	1) Check power supply, lead and wiring 2) If power supply is ok, Check internal electrical connections 3) If electrical connections are ok, Check compressor windings 4) If compressor is ok, Change compressor control module
E 04	Compressor speed low	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
E 05	Controller overheated	1) Is ambient temperature very high? 2) Ensure ventilation is not blocked 3) turn off device for 1 hour then restart
E 06	NTC open circuit	1) Check NTC connection 2) If connection is ok, replace NTC
E 07	NTC short circuit	1) Check NTC for short circuit 2) If NTC is ok, replace compressor control module
E 09	Valve faulty (Dual zone only)	1) Check if error disappears automatically within 30 seconds (can be false triggered by zone switching at the same time as error detection) 2) Check connections at valve and compressor control. 3) If connections are ok, check cooling performance of each zone, 4) If at least one zone is not cooling, activate valve and check if valve is receiving power. 5) If valve is not receiving power, replace module. 6) If valve is receiving power, visually check for blockages 7) If no sign of blockages, replace valve.
E 10	Communication error between display and controller	1) Check connections at compressor control module and Display PCB 2) If connections are ok, Replace display PCB 3) If Replacing the display PCB hasn't fixed the issue, replace Compressor control module

COMPRESSOR START FAILURE

3 Flash error – CF, CFF, CFX

Warning 33 – CFX3, CFX5

Error 03 – CFX2

1. INSUFFICIENT DC SUPPLY VOLTAGE

The most common cause of compressor start failure is insufficient supply voltage.

Common causes of poor supply voltage include:

- Undersized cabling
- Loose connections
- Faulty battery
- Faulty lead

Note:

You may read a suitable voltage at the fridge however once the compressor attempts to start (then creating a load) the voltage can drop significantly then causing the compressor to stall and display warning 33.

2. FAULTY CONTROL UNIT

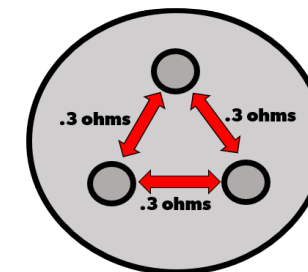
If you have established sufficient supply voltage it is likely that the control unit is faulty.

Before replacing the control unit, verify that all connections are secure, particularly the connection between the control unit and the compressor windings.



3. FAULTY COMPRESSOR

Check resistance of compressor windings, all windings should read .3 ohms, if a winding is open circuit the compressor is faulty.



BATTERY PROTECTION SETTING

Setting the battery protection level to **Low** allows the refrigerator to operate at a lower supply voltage.

For example, if the battery protection is set to **High (12.6v protection)** and the refrigerator is supplied with only **12.2 V**, this will prevent the compressor from starting and result in a **Compressor start failure** error.

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Low Medium/High

		Low	Medium	High
12 V	OFF	10.1 V	11.2 V	11.8 V
	ON	11.4 V	12.2 V	12.6 V
24 V	OFF	21.5 V	24.1 V	24.6 V
	ON	23.0 V	25.3 V	26.2 V

619 CONFIG APP

When is configuration required?

- If the original Compressor Controller is replaced.
- If the "Reset type ID" has been chosen in the CFX3 fridge/freezer.

Note: configuration is not required if replacing only the display PCB.

What is the purpose?

To align the electronics with one of three CFX3 configuration types

- Single Zone
- Icemaker
- Dual Zone

Preparation

1. Search "619 Config" on App Store or Google Play and download to your chosen device.
2. Ensure your device Bluetooth is switched on, 'Forget' any other Bluetooth connections on your device.
3. Allow notifications & permissions on your device for the app.
4. Ensure CFX3 fridge/freezer is not already paired to an in-range device other than the one to be used for configuring.
5. Ensure CFX3 fridge/freezer Bluetooth is turned on before & starting the configuration process. **(IMPORTANT)**



619 CONFIG APP

When replacing a compressor Control module

1. Switch CFX3 fridge/freezer Bluetooth **ON** before replacing compressor controller or selecting the "Reset type ID" (Otherwise display showing QR code will be frozen, and configuration not possible)
2. Disconnect the power and replace the compressor control module.
3. Once control module is replaced, double check that Bluetooth is switched on and then reset type ID .

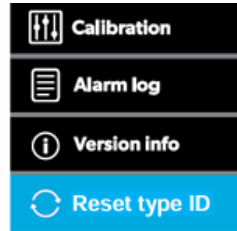
S/Nos between
9xxxxxxx - 02904016

S/Nos from
02904017 -

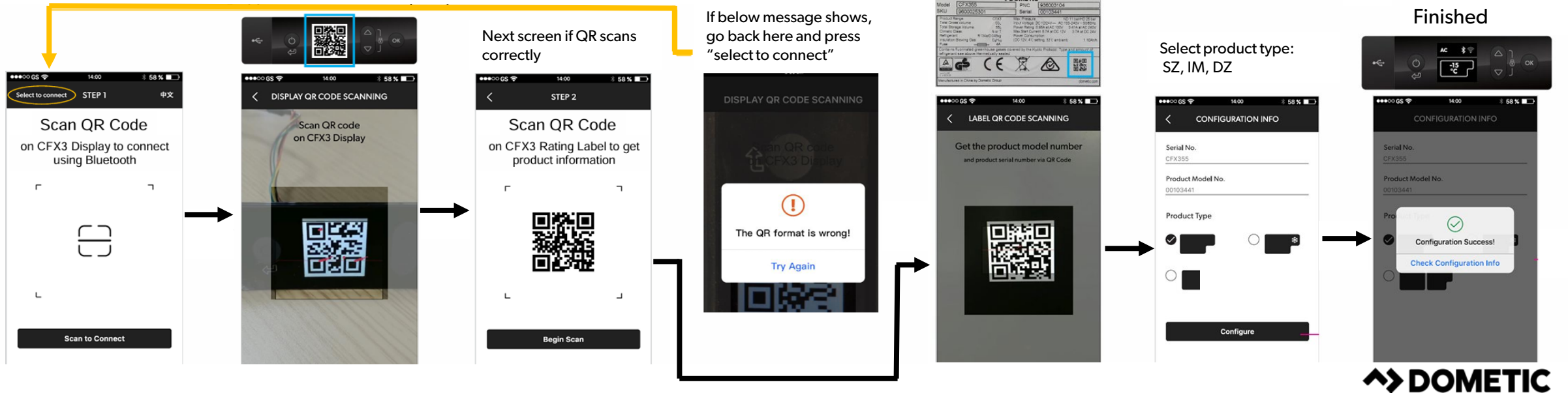
Press & Hold
 + 
for 3 sec

OR

Press & Hold
 + 
for 5 sec



4. Open the 619 Config app and follow the screen prompts as follows



619 CONFIG APP

Display QR Code Frozen

If the compressor controller is replaced with Bluetooth originally switched off, the 619 Config app will not work and the QR code will be frozen in the display, There are two methods to recover from this condition

Method 1:

If you have access to another CFX3 fridge/freezer

CFX3(A) = The frozen display unit,

CFX3(B) = Another CFX3 with display working ok (can be any type, SZ, DZ or 55IM)

1. Turn on CFX3(B) Bluetooth
2. Power off both CFX3(A) and CFX3(B) & disconnect power cables
(Failure to power off whilst removing/refitting displays may permanently damage the display electronics)
3. Mark one of the displays (for identification after removal), disconnect & remove both displays
4. Fit display from CFX3(B) to CFX3(A), connect power cable & power on CFX3(A)
(Bluetooth on/off is stored in the display, so Bluetooth now on.)
5. Perform the 619 Config app process on CFX3(A).
6. Fit display from CFX3(A) to CFX3(B), connect power cable & power on CFX3(B).
(Type data (SZ, DZ or 55IM) & S/No stored in the compressor controller, so CFX3(B) will work immediately so no need to run the Config app, note Bluetooth now off)
or
If either/both displays are scratched & belong to customers, power off both CFX3's and switch the displays back to original CFX3.

Method 2:

If you do not have access to another CFX3 fridge/freezer

1. Power off the CFX3, remove the new compressor controller and refit the original compressor controller.
2. Power on. The QR code should now be gone.
3. Turn on the Bluetooth.
4. Power off, refit new compressor controller, power on and run the 619 Config app process.

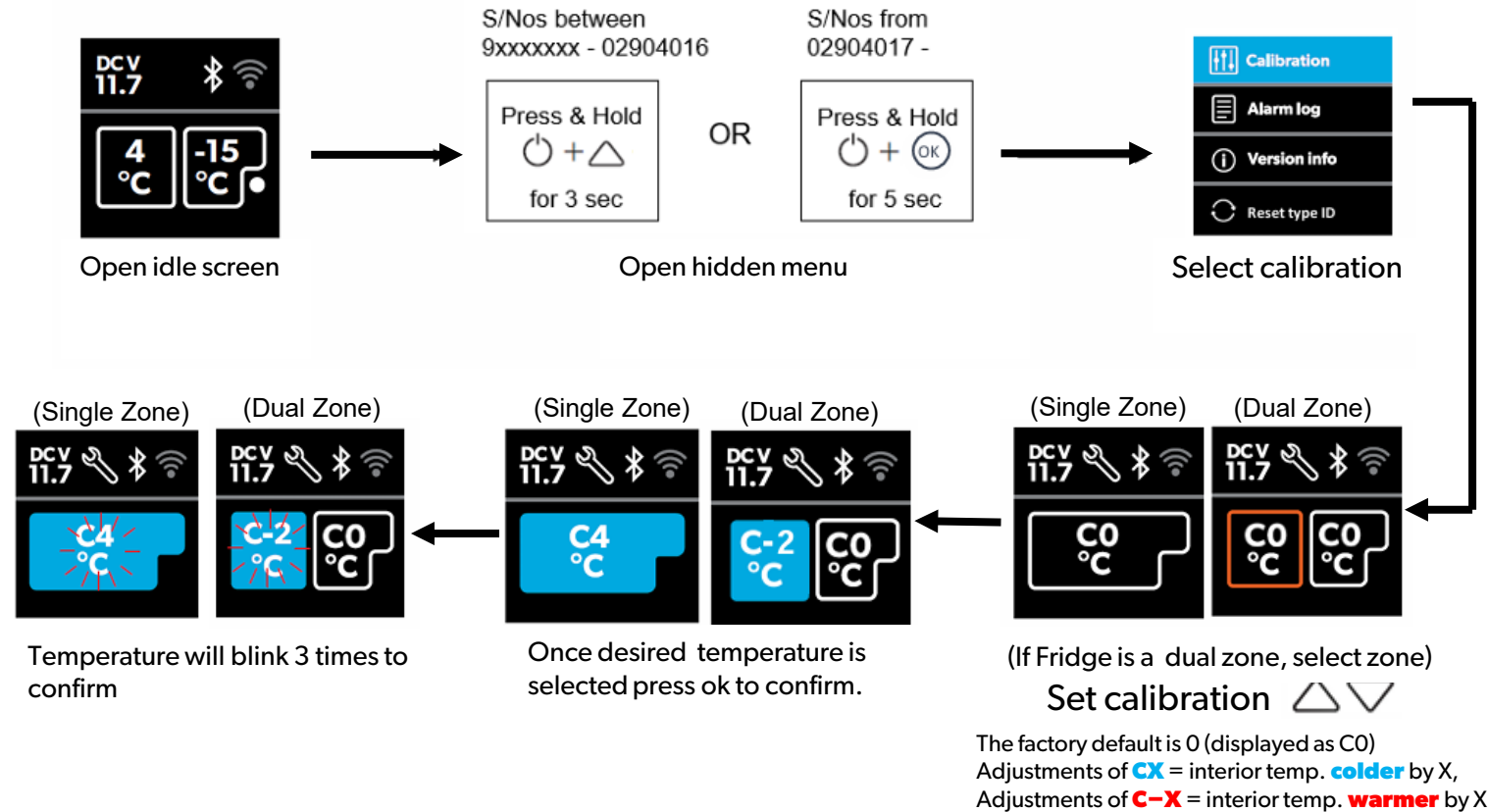
CFX3 & CFX5 TEMPERATURE CALIBRATION

Temperature calibration aka temperature offset changes the cut in/out temperature to achieve an accurate temperature reading.

If your display is reading 4°C when your actual cabinet cycling temperature is 0°C, you will calibrate the temperature to C-4°C, so display temperature and actual temperature are the same.

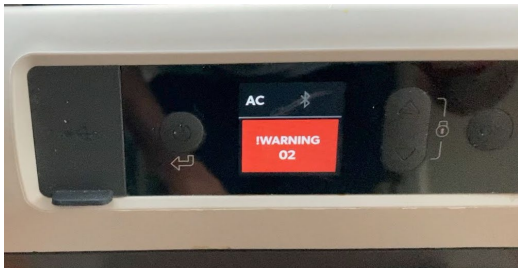
Temperature calibration can only influence operation when the fridge is cycling.

If the fridge is not cycling, making the calibration setting "colder" will not resolve the issue. Check for possible causes such as a faulty condenser fan, restricted airflow, insufficient ventilation, or a blocked condenser.



FAULTY THERMISTOR SYMPTOMS

ERROR MESSAGES



CFF

Err 01 (SZ) / Er 01 (DZ)	NTC open circuit SZ & DZ LARGE
Err 01 (SZ) / Er 11 (DZ)	NTC open circuit DZ SMALL
Err 02 (SZ) / Er02 (DZ)	NTC short circuit SZ & DZ LARGE
Err 12 (SZ) / Er12 (DZ)	NTC short circuit DZ SMALL

CFX & CF

Err 1	Thermistor circuit closed
Err 2	Thermistor circuit open

CFX2

E 06	Thermistor circuit open
E 07	Thermistor circuit closed

CFX3, CFX5

! Warning 01	Error_Thermostat Faulty	NTC open circuit SZ & DZ Large
! Warning 11	Error_Thermostat Faulty	NTC open circuit DZ Small
! Warning 02	Error_Thermostat Faulty	NTC short circuit SZ & DZ Large
! Warning 12	Error_Thermostat Faulty	NTC short circuit DZ Small

FAULTY THERMISTOR SYMPTOMS

INCORRECT TEMPERATURE DISPLAYED



1. The portable refrigerator may display an abnormal temperature.

Eg the fridge is running, and the display is reading 58 °C



2. The refrigerator is set to a temperature above freezing ($>0^{\circ}\text{C}$), and the cabinet temperature is freezing.

Eg the fridge is set/displaying 3°C and the cabinet temperature is -22°C

THERMISTOR RESISTANCES

Temperature °C	Resistance kΩ
0°C	27.3720
1°C	26.2161
2°C	25.1155
3°C	24.0673
4°C	23.0685
5°C	22.1167
6°C	21.2095
7°C	20.3444
8°C	19.5193
9°C	18.7321
10°C	17.9810
11°C	17.2641
12°C	16.5796
13°C	15.9260
14°C	15.3016
15°C	14.7050
16°C	14.1349
17°C	13.5899
18°C	13.0688
19°C	12.5705
20°C	12.0938
21°C	11.6377
22°C	11.2011
23°C	10.7833
24°C	10.3832
25°C	10.0000
26°C	9.6330
27°C	9.2813
28°C	8.9443
29°C	8.6212
30°C	8.3115

EMERGENCY OVERRIDE

Previous models CF and CFX had an emergency override switch installed; This switch was installed to bypass a faulty main board incase of an emergency. In this mode, the compressor runs continuously, ignoring all thermostat and control panel settings.

This function can be mistaken for a faulty thermistor (especially by consumers).

ON



OFF



A white motorhome is driving away from the viewer on a winding dirt road that cuts through a vast, mountainous landscape. The terrain is covered in dry, golden-brown grass and low-lying shrubs. In the distance, rugged mountains rise under a clear sky. The motorhome has a yellow license plate that reads 'W588 NES' and is carrying gear on its roof rack. The overall scene conveys a sense of adventure and outdoor travel.

 **DOMETIC**